

Shapley credit assignment for Dreamer in terminal-reward tasks

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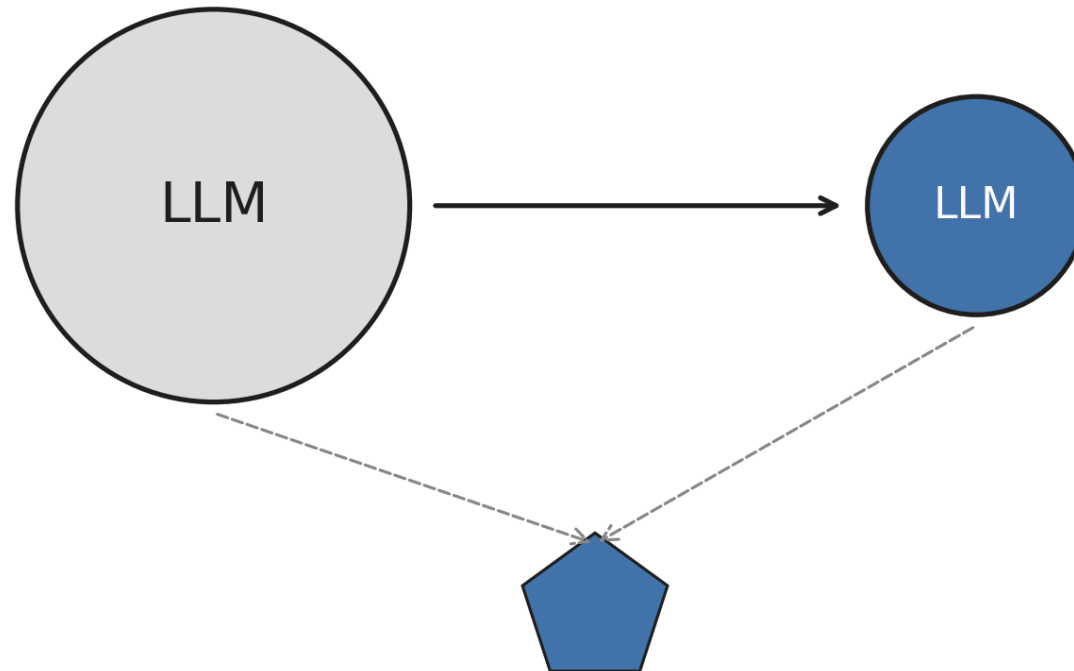
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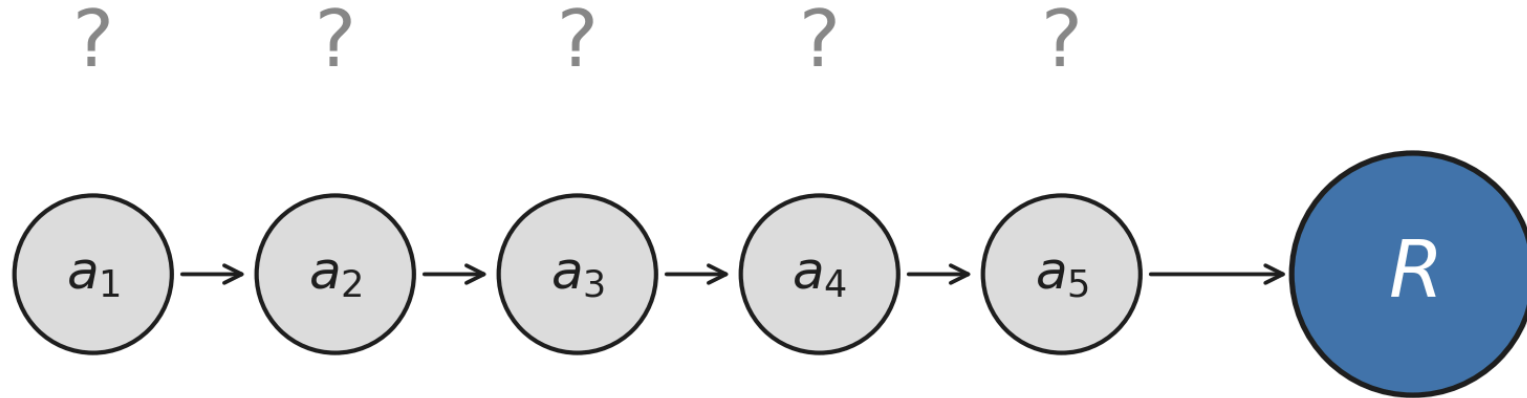
Master program in Applied Informatics

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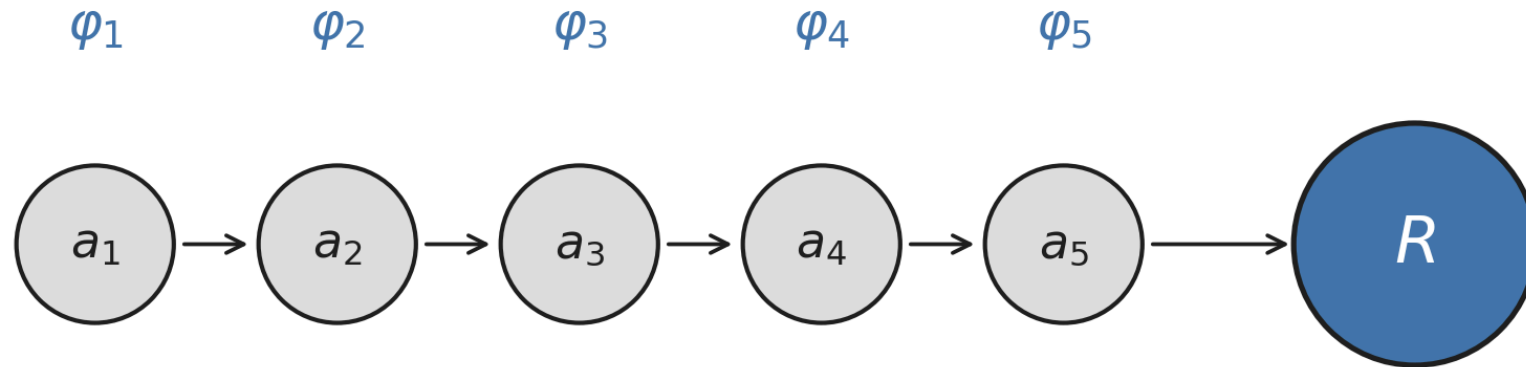
Main motivation



Setting: RL with terminal reward

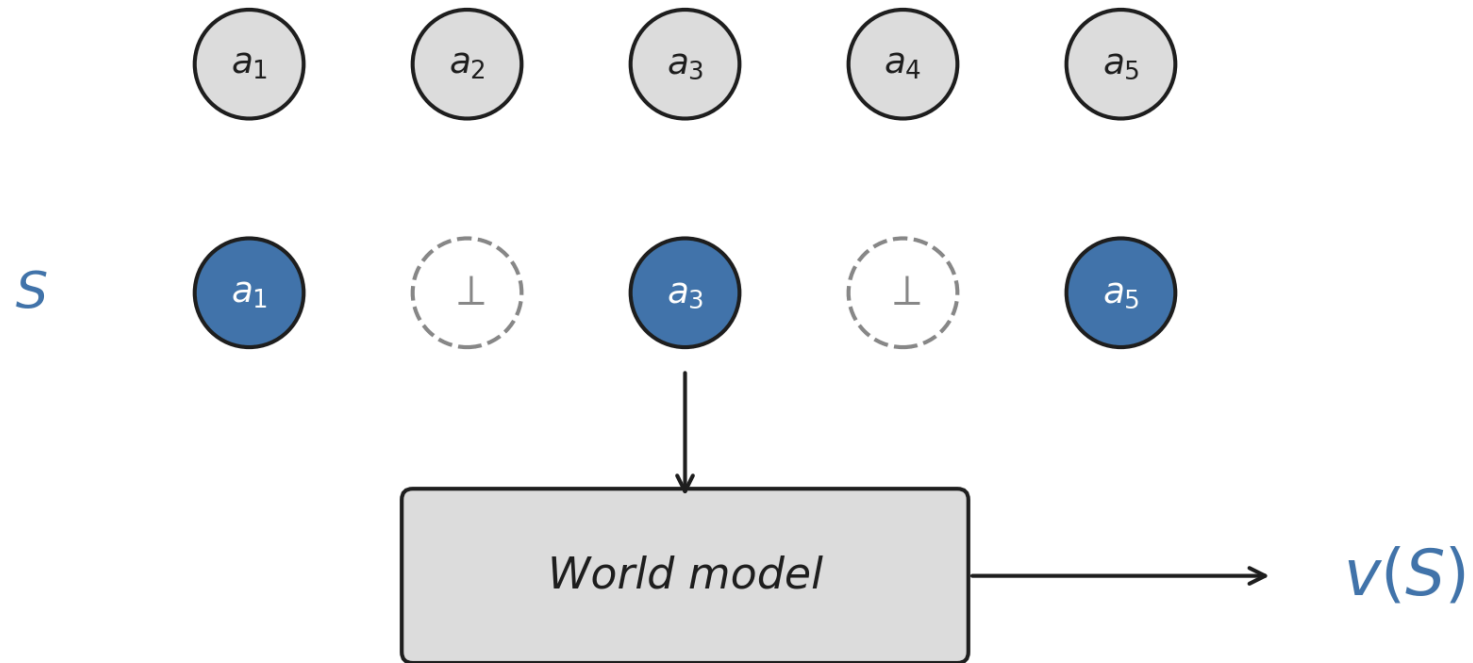


Tool 1: Shapley decomposition

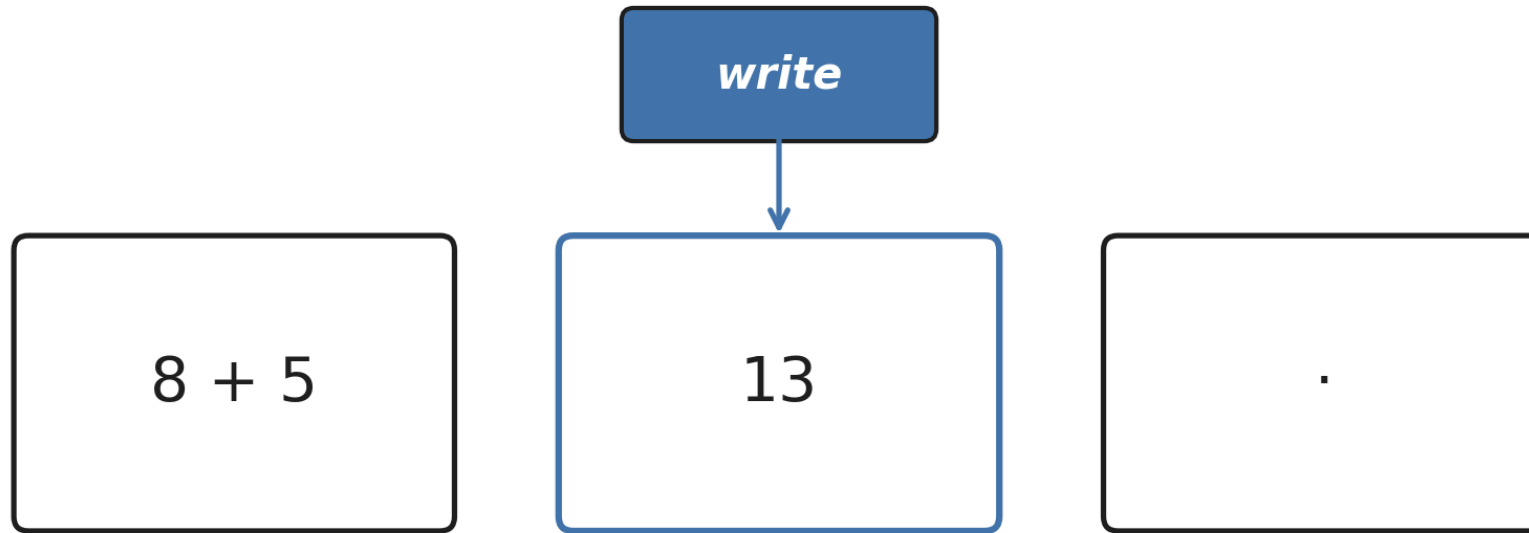


$$\sum_i \varphi_i = R$$

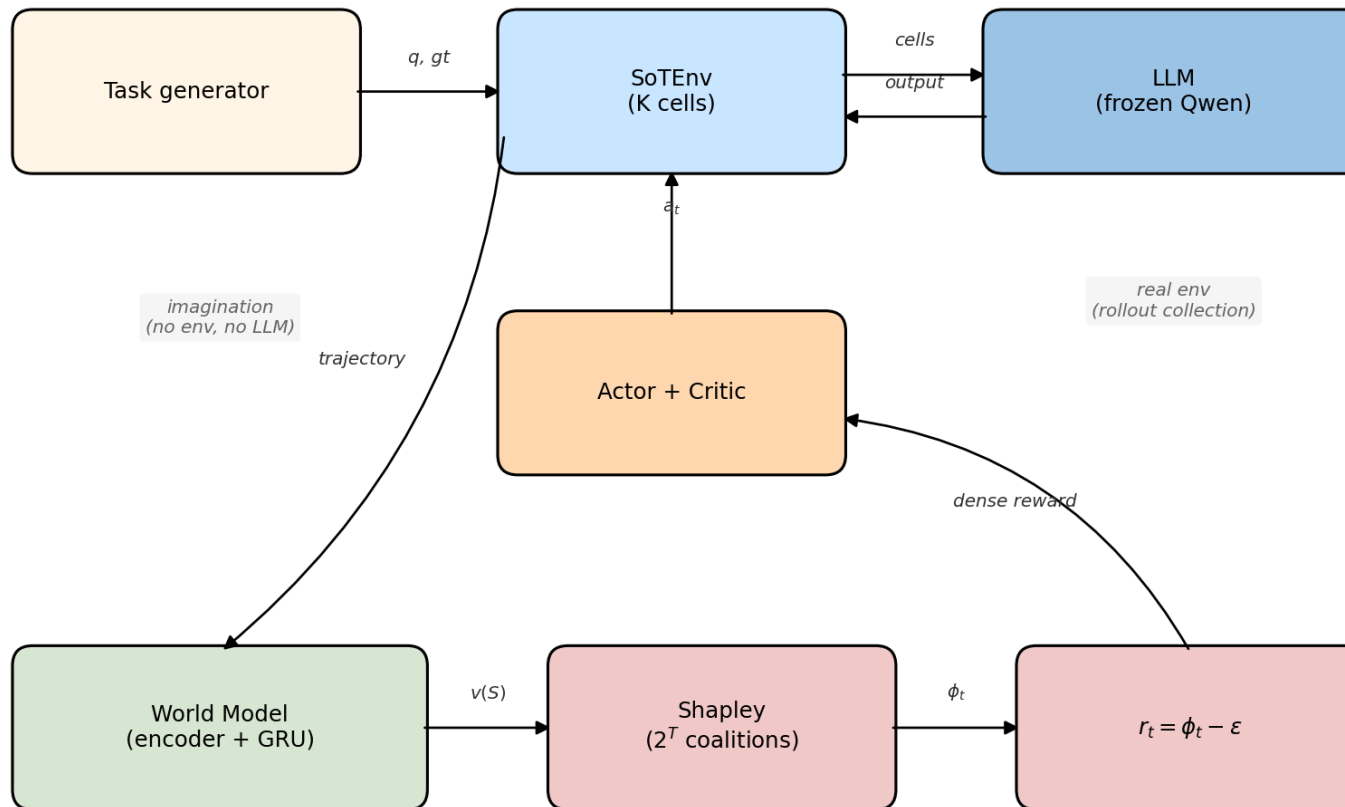
Tool 2: world model as oracle $v(S)$



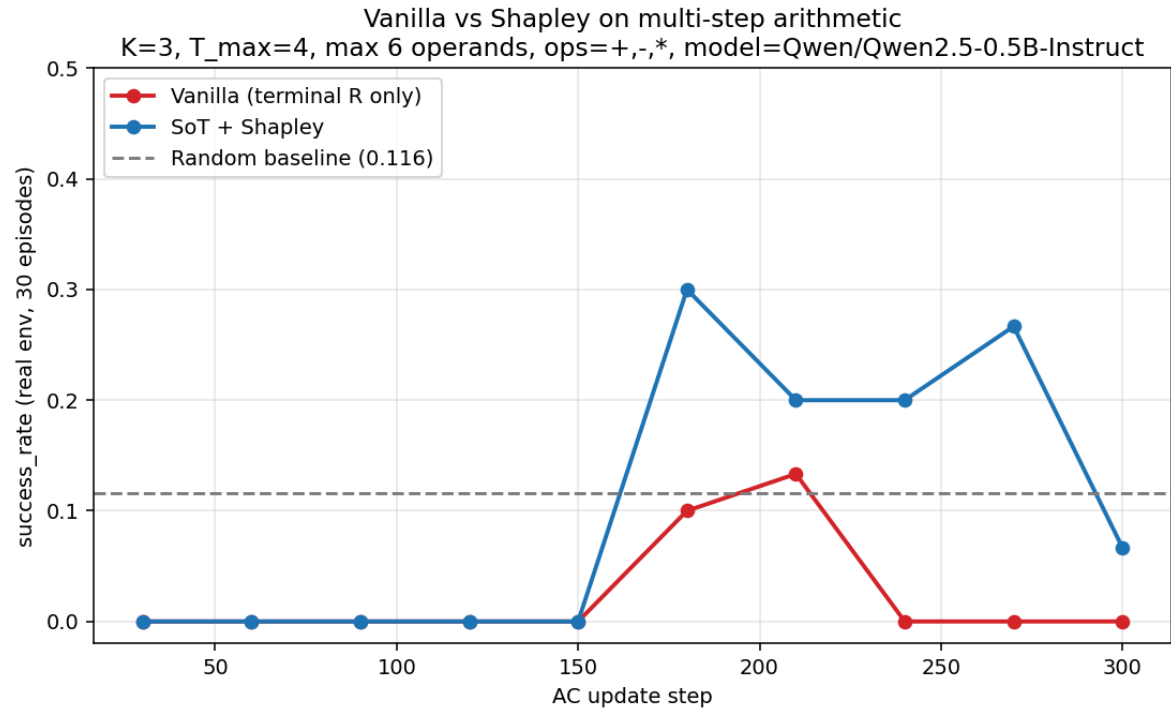
Agent: cell-based working memory



Architecture



First experiment

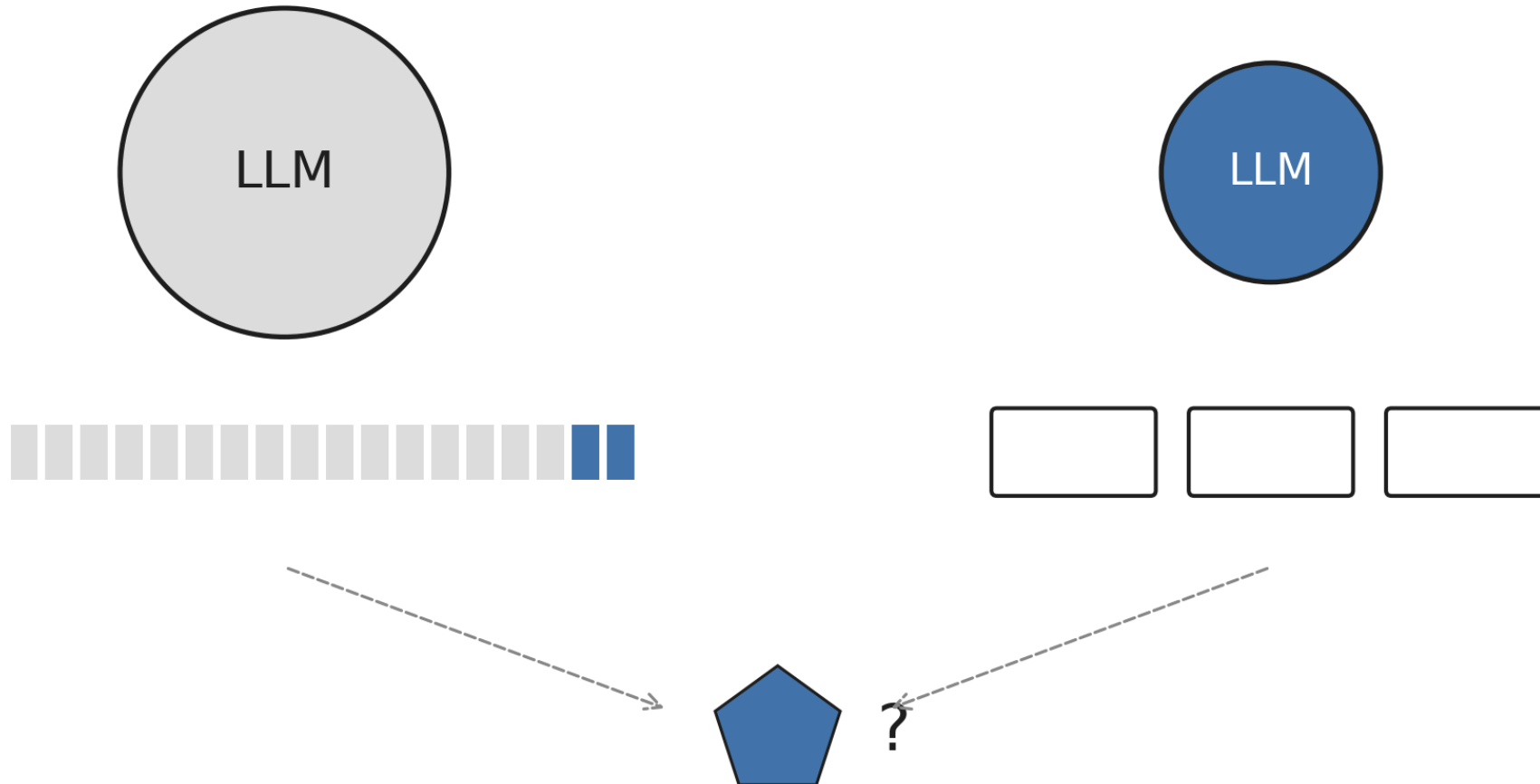


- Random — 0.116
- Vanilla — 0.133
- **Shapley — 0.300**

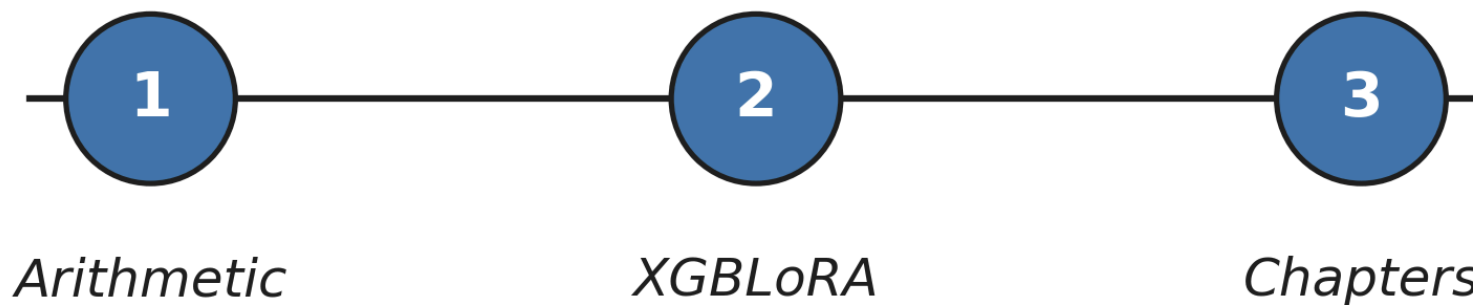
Back to the LLM: CoT bloat



Long-term hypothesis: smaller LLM with cells



Near-term steps



Next: compare against competing methods

AgeMem

2026

MemAct

2025

RUDDER

2019

Summary

GitHub: github.com/51nn3r/dp

Thesis: [thesis/en/main.pdf](#) · [thesis/sk/main.pdf](#)

Thank you